SOI Cartographer Challenge

Submission Execution Guide

Apratim Das

July 16, 2026

Project Overview

This submission contains a complete ROS2 Jazzy autonomous multi-robot exploration solution developed for the SOI Cartographer Challenge.

The project performs:

- Dual robot exploration
- Online occupancy grid mapping
- Frontier detection
- Frontier allocation
- A* path planning
- Pure Pursuit based path following
- Collision avoidance
- Global map merging

Project Folder

The workspace should have the following structure:

```
ros2_ws/
  src/
   Apratim_Solution/
   challenge_bridge/
   team_solution/
  build/
  install/
  log/
```

System Requirements

- Ubuntu 24.04 (WSL2 or Native)
- ROS2 Jazzy
- Gazebo Harmonic
- Python 3.12
- Colcon

Build Instructions

Open a terminal and execute:

```
cd ~/ros2_ws

rm -rf build install log

source /opt/ros/jazzy/setup.bash

colcon build --symlink-install

source install/setup.bash
```

The build should finish without errors.

Launching the Competition Environment

Terminal 1

```
cd ~/ros2_ws

source /opt/ros/jazzy/setup.bash
source install/setup.bash

ros2 launch challenge_bridge competition.launch.py
```

Wait until Gazebo opens and both robots are spawned.

Launching the Solution

Open a second terminal.

```
cd ~/ros2_ws

source /opt/ros/jazzy/setup.bash
source install/setup.bash

ros2 run Apratim_Solution solution_node
```

The solution node should initialize successfully.

Verification Commands

To verify that the system is running correctly:

Nodes

```
ros2 node list
```

Expected nodes include:

- solution_node
- robot_1_bridge
- robot_2_bridge
- global_bridge

Topics

```
ros2 topic list
```

Important topics:

- /robot_1/scan
- /robot_2/scan
- /robot_1/odom
- /robot_2/odom
- /robot_1/cmd_vel
- /robot_2/cmd_vel
- /global_map

Check Laser Data

```
ros2 topic echo /robot_1/scan --once
```

Check Odometry

```
ros2 topic echo /robot_1/odom --once
```

Check Velocity Commands

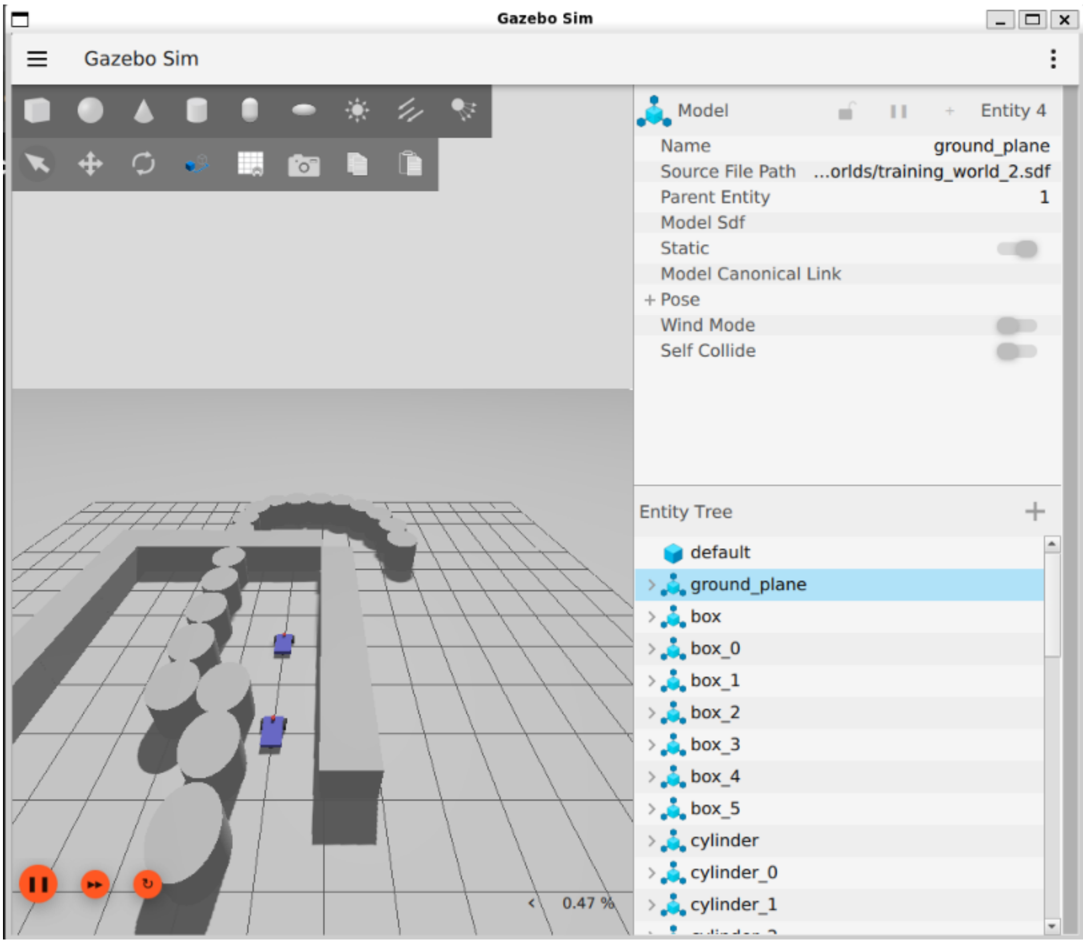
```
ros2 topic echo /robot_1/cmd_vel
```

Expected Behaviour

After launching:

1. Gazebo loads successfully.
2. Two robots spawn.
3. The solution node starts.
4. Robots autonomously explore the environment.
5. Occupancy map grows over time.
6. Frontier exploration continues until completion.

Simulation Screenshot



Troubleshooting

If rebuilding is required:

```
cd ~/ros2_ws

rm -rf build install log

source /opt/ros/jazzy/setup.bash

colcon build --symlink-install

source install/setup.bash
```

If ROS environment variables are missing:

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
```

Submission

The provided solution package contains all required source code necessary to build and execute the project using the commands described above.

End of Document